Team name - Tokenizers

Project Title - Cartoon img captioning

1. Ventura et al. (2025)

Ventura, L., Schmid, C. & Varol, G. Learning Text-to-Video Retrieval from Image Captioning. Int J Comput Vis 133, 1834–1854 (2025). https://doi.org/10.1007/s11263-024-02202-8

Ventura, Schmid, and Varol (2025) propose a framework that leverages knowledge from image captioning to improve text-to-video retrieval tasks. The study utilizes large-scale datasets such as MSCOCO for image-caption pairs and MSR-VTT for video-text alignment. Their approach is based on transfer learning, where models trained on image captioning are adapted to understand video content using cross-modal embeddings and transformer-based architectures. This method significantly reduces the dependency on large annotated video datasets and improves retrieval performance. However, the model struggles with capturing temporal relationships in videos and relies heavily on caption quality. For a cartoon image captioning project, this work suggests that transfer learning from real-world image datasets can be highly beneficial. To overcome its limitations, we can fine-tune the model on cartoon-specific datasets and incorporate style-aware features to better handle exaggerated and non-realistic visual elements.

2. Andrade López & Flexer

ZERO-SHOT OPEN SET OBJECT DETECTION IN MUSIC ALBUM COVERS Selene X. ANDRADE LÓPEZ1 and Arthur FLEXER(arthur.flexer@jku.at) 1 1 Institute of Computational Perception, Johannes Kepler University Linz, Austria

Andrade López and Flexer explore zero-shot open-set object detection in music album covers, focusing on identifying objects that are not present in the training dataset. The study likely uses curated album cover datasets along with pretrained vision-language models such as CLIP. Their technique relies on aligning textual and visual embeddings, enabling the model to recognize unseen objects using semantic descriptions. This approach is particularly effective for artistic and unconventional images, making it highly relevant for cartoon-like data. The major advantage is its ability to generalize without requiring extensive labeled datasets. However, the model may produce less accurate predictions and is sensitive to prompt design. In the context of cartoon image captioning, this approach can be leveraged by integrating CLIP-based embeddings to improve generalization. To overcome its drawbacks, combining zero-shot capabilities with fine-tuning on cartoon datasets and using well-structured prompts can enhance accuracy.

3. Siriborvornratanakul & Rittikulsittichai (2026)

Siriborvornratanakul, T., Rittikulsittichai, S. Advancing the generation and integration of traditional motifs through AI-based techniques. Discov Artif Intell 6, 100 (2026). https://doi.org/10.1007/s44163-025-00642-w

Siriborvornratanakul and Rittikulsittichai (2026) focus on generating and integrating traditional artistic motifs using AI-based techniques. The study uses datasets consisting of cultural and artistic patterns and applies generative models such as GANs and diffusion models to learn stylistic representations. Their approach emphasizes preserving artistic characteristics while generating new designs, making it highly suitable for stylized domains. The main advantage lies in its ability to capture and reproduce complex artistic styles. However, the model may lose semantic meaning and can be difficult to evaluate objectively. For a cartoon image captioning project, this work highlights the importance of incorporating style-awareness into the model. By adding style descriptors (e.g., cartoon, anime, sketch) into captions and training the model to recognize artistic patterns, we can improve caption quality. To address limitations, combining style learning with semantic understanding ensures both accuracy and creativity.

4. Xu et al. (2025) – PointLLM-V2

R. Xu et al., "PointLLM-V2: Empowering Large Language Models to Better Understand Point Clouds," in IEEE Transactions on Pattern Analysis and Machine Intelligence, doi: 10.1109/TPAMI.2025.3590784. keywords: {Point cloud compression;Three-dimensional displays;Training;Data models;Solid modeling;Large language models;Data collection;Pipelines;Benchmark testing;Measurement;Computer Vision;Natural Language Processing;Multi-Modal LLM;3D Understanding;Point Cloud}

Xu et al. (2025) introduce PointLLM-V2, a multimodal framework designed to enable large language models to understand 3D point cloud data. The model is trained on datasets such as ShapeNet and ScanNet and integrates geometric data with language representations. The technique demonstrates how multimodal learning can enhance understanding across different data types by combining spatial and textual features. Its key advantage is strong reasoning capability across modalities, enabling detailed and context-aware descriptions. However, the model is computationally expensive and complex, and its direct application to 2D images is limited. For cartoon image captioning, the main takeaway is the importance of multimodal feature fusion. By combining visual features with textual embeddings, we can build a more robust captioning system. To mitigate complexity, lightweight multimodal architectures such as BLIP can be used instead of heavy 3D models.

5. Soni et al. (2025)

J. Soni, H. Upadhyay, P. P. A. Victor and S. Tripathi, "Evaluating Generative AI Models for Image-Text Modification," in IEEE Access, vol. 13, pp. 40703-40729, 2025, doi:

10.1109/ACCESS.2025.3547677. keywords: {Measurement;Diffusion models;Context modeling;Benchmark testing;Feature extraction;Translation;Object recognition;Image segmentation;Data models;Text to image;Generative AI;image editing;diffusion models;human-evaluation;auto rating metrics}

Soni et al. (2025) evaluate generative AI models for image-text modification, focusing on how effectively models can edit images based on textual input. The study uses image-text datasets and benchmarks diffusion models using both automatic metrics and human evaluation. Their approach highlights the importance of strong alignment between visual and textual modalities and introduces evaluation techniques such as BLEU, METEOR, and human judgment. The advantages include improved image-text consistency and robust evaluation frameworks. However, diffusion models are computationally expensive and evaluation metrics may not fully capture human perception. In a cartoon image captioning project, this work emphasizes the need for proper evaluation strategies. By incorporating both automatic metrics and human feedback, we can better assess caption quality. To overcome limitations, combining efficient models with balanced evaluation methods ensures both performance and practicality.

Our Proposed Approach (Cartoon Image Captioning)

In our cartoon image captioning project, we propose a hybrid approach that combines transfer learning, zero-shot learning, and multimodal understanding to generate accurate and context-aware captions for stylized images. Initially, a pretrained model such as CLIP is used to extract rich visual-text embeddings and handle unseen objects through zero-shot learning. This is followed by a transformer-based caption generation model (e.g., BLIP or a custom encoder-decoder architecture) fine-tuned on both real-world datasets like MSCOCO and a curated cartoon image dataset to adapt to stylized features. Additionally, inspired by style-based generation techniques, we incorporate style-aware tokens (e.g., cartoon, anime) to improve caption relevance. The model leverages multimodal fusion to combine visual features and textual context effectively. For evaluation, both automatic metrics (BLEU, METEOR) and human evaluation are used to ensure quality and coherence. This approach addresses limitations of existing methods by improving generalization, handling artistic styles, and maintaining semantic accuracy in cartoon image captioning.